Highlights

Adapting Geospatial Foundation Models for Urban Socioeconomic Prediction: A Paired Multimodal Evaluation on *CityLens*

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- PRITHVI-EO-2.0+LoRA leads all satellite encoders on four CityLens tasks despite RGB-only input, outperforming DINOv2 by up to 0.291 R^2 points.
- A shared-split paired protocol reveals fusion improves only building height ($R^2 = 0.850 \pm 0.032$); satellite alone dominates GDP and healthcare.
- GDP street-branch R^2 flips sign across two seeds ($+0.47 \rightarrow -0.23$), demonstrating the hazard of single-seed EO benchmarking.
- Population density is unrecoverable from RGB imagery at this spatial scale across all modalities and fusion strategies.

Adapting Geospatial Foundation Models for Urban Socioeconomic Prediction: A Paired Multimodal Evaluation on *CityLens*

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ABSTRACT

Predicting neighbourhood-scale urban indicators from imagery is frequently studied under evaluation protocols that confound model quality with data composition—modality-specific validation splits and single-seed reporting make it difficult to determine whether observed performance differences reflect genuine signal or statistical artefacts. This paper addresses both challenges simultaneously. We introduce a *shared-split paired evaluation framework* that fixes identical held-out region identifiers across satellite, street-view, and fusion branches for each task and random seed, eliminating the modality-split confound by construction. Within this framework, we adapt the PRITHVI-EO-2.0 geospatial foundation model to RGB-only input via a learnable 1×1 convolutional channel adapter and parameter-efficient LORA fine-tuning, and benchmark it against ResNet-50, DINOv2, and CLIP baselines on four *CityLens* urban indicator regression tasks: GDP proxy, healthcare accessibility, building height, and population density. PRITHVI-EO-2.0+LORA achieves the highest R^2 among all satellite encoders on every task despite receiving only three of the six spectral channels it was pretrained to expect. Paired two-seed results ($s \in \{42, 43\}$) yield three key findings: (i) building-height fusion is the only task where multimodal combination helps, achieving $R^2 = 0.850 \pm 0.032$ versus satellite-only 0.811 ± 0.083; (ii) GDP street-branch R^2 flips sign between seeds, turning a seemingly viable modality into one worse than a constant-mean predictor; and (iii) population density resists prediction from RGB imagery at this spatial scale across all configurations. These results demonstrate that more modalities do not always help and that single-seed evaluation can produce directionally wrong conclusions. The evaluation framework—shared splits, multi-seed paired reporting, and structured failure documentation—is directly applicable to any multimodal Earth observation foundation model benchmark.

Plain language summary

Machine learning models can estimate city-scale statistics—economic output, healthcare access, building height—from satellite and street-level photographs. However, different research groups often compare models unfairly: they test each modality on different subsets of cities and report results from a single random data split. We introduce a fairer evaluation method that tests all models on exactly the same cities and repeats experiments at multiple random seeds. Using this approach, we show that a NASA–IBM Earth observation foundation model, originally designed for multispectral satellite data, outperforms general-purpose image models even when it receives only ordinary colour photographs. Combining satellite and street photographs improves predictions for building height only; for GDP and healthcare access, the satellite model alone performs better. Population density cannot be predicted from photographs at this scale. Critically, we show that the relative ranking of satellite and street models can reverse between two random seeds for GDP prediction—a concrete demonstration that single-seed evaluation can produce misleading conclusions.

1. Introduction

More than half the world's population lives in cities, yet measuring urban socioeconomic conditions at the spatial resolution needed for neighbourhood-scale policy remains difficult. National censuses, administrative health records, and household expenditure surveys are expensive, slow to update, and geographically uneven—systematically underserving the fastest-growing regions in the Global South (Jean, Burke, Xie, Davis, Lobell and Ermon, 2016). Machine learning

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applied to geospatial imagery promises to close this gap: satellite and street-view photographs encode rich visual signatures of economic activity, built-environment morphology, and infrastructure that can be decoded into spatially contiguous socioeconomic estimates (Liu, Pang, Zhang, Ouyang, Zhang, Feng, Li and Hui, 2026; Fan, Zhang, Loo and Ratti, 2023; Gebru, Krause, Wang, Chen, Deng and Fei-Fei, 2017).

This promise has catalysed a wave of modelling research, but the field now faces a credibility challenge rooted in evaluation practice rather than model architecture. Two pervasive protocol deficiencies undermine the reliability of published results. The first is *modality-specific validation splits*: when satellite and street-view models are evaluated on independently drawn held-out folds—which may not even overlap geographically—the resulting R^2 values conflate modality signal with split composition. A validation set over-representing wealthy, well-imaged cities inflates satellite R^2; one with atypical street-view coverage inflates street R^2. The second is *single-seed reporting*: a model's score at one random seed may reflect a statistically favourable draw rather than genuine superiority, yet most published comparisons treat a single number as definitive. We term the combination of these issues the *modality-split confound* and demonstrate empirically that it can reverse the apparent ordering of satellite and street models on GDP prediction.

Earth observation (EO) foundation models introduce a new axis along which to study this problem. PRITHVI-EO-2.0 (IBM Research and NASA-IMPACT, 2025), developed by IBM Research and NASA-IMPACT, is a Vision Transformer pretrained on large-scale multitemporal, multispectral Harmonized Landsat-Sentinel (HLS) imagery using a masked autoencoder objective. Its architecture ingests six spectral bands spanning the visible through shortwave-infrared range, raising a natural question: when only RGB imagery is available—as is typical in urban sensing applications including *CityLens*—does the spatial knowledge encoded during EO pretraining still confer advantage over general-purpose models such as DINOv2 (Oquab, Darcet, Moutakanni et al., 2024) and CLIP (Radford, Kim, Hallacy et al., 2021), which were trained on natural photographs? Answering this question requires both a principled adaptation strategy for the spectral mismatch and an evaluation protocol that does not confound model quality with split composition.

Paired evaluation framework. We address both challenges jointly. For each task and random seed, we generate a shared split file that fixes validation region identifiers across all three modality branches—satellite, street-view, and fusion—so every model variant is scored on exactly the same held-out cities (Equation 14). This eliminates the modality-split confound by construction: observed R^2 differences reflect only differences in visual signal, not artefacts of split composition. We report results across two random seeds ($s \in \{42, 43\}$) to characterise the variance of branch-level orderings, moving beyond point estimates toward distributional claims.

On the modelling side, we adapt PRITHVI-EO-2.0 to RGB-only input via a learnable 1×1 convolutional channel adapter (Equation 5) that maps three channels to the six expected by the backbone, combined with parameter-efficient LoRA (Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang and Chen, 2022) fine-tuning of frozen attention layers (Equation 7). This preserves EO spatial priors while permitting task-specific adaptation on the few hundred training examples available per *CityLens* target. We benchmark this adapted Prithvi against ResNet-50 (He, Zhang, Ren and Sun, 2016) and DINOv2 (Oquab et al., 2024) satellite encoders, as well as ResNet-50, CLIP (Radford et al., 2021), and DINOv2 street encoders with mean or attention multi-view pooling (Equations 9–**??**), evaluating late and gated fusion architectures (Equations 11–12).

Contributions.

C1. Shared-split evaluation protocol. A reproducible mechanism that fixes validation region IDs across satellite, street-view, and fusion branches via explicit JSON files keyed by task, seed, and split tag, eliminating the modality-split confound that undermines cross-modal comparison in prior work (Section 4.6).

C2. Systematic RGB adaptation of an EO foundation model. The first controlled assessment of PRITHVI-EO-2.0+LoRA with a learnable channel adapter on *CityLens*, benchmarked against CNN and ViT baselines across satellite, street-view, and fusion branches (Sections 4.2–4.5).

C3. Task-dependent fusion value. Empirical evidence that multimodal fusion is not universally beneficial: it improves building height prediction, degrades GDP and healthcare prediction, and fails entirely on population density—challenging the implicit assumption that more modalities always help (Section 5).

C4. Multi-seed analysis with structured failure documentation. Two-seed paired results with mean ± std reporting, alongside five documented failure modes including a catastrophic baseline ($R^2 \approx -4.0$) and a sign flip in street-branch GDP R^2 across seeds—concrete evidence of single-seed evaluation hazards (Section 6.5).

2. Related Work

2.1. Geospatial Foundation Models

Foundation models pretrained on large Earth observation (EO) corpora have rapidly extended the self-supervised paradigm from natural language processing to remote sensing. The central insight is that the spatial, spectral, and temporal structure of satellite imagery provides rich self-supervisory signal: learning to reconstruct masked patches across time and spectrum yields general-purpose representations of land cover, urban morphology, and environmental dynamics without labelled data.

PRITHVI-EO-2.0 (IBM Research and NASA-IMPACT, 2025) exemplifies this approach, combining a Vision Transformer backbone with 3D spatiotemporal patch embeddings pretrained on multitemporal HLS data via masked autoencoding. It has demonstrated strong transfer to segmentation and change-detection tasks. SatMAE (Cong, Khanna, Meng et al., 2022) takes a complementary approach, treating each spectral band as an independent temporal token to learn cross-band correlations without imposing fixed spectral ordering. SpectralGPT (Hong, Zhang, Li et al., 2024) extends pretraining to full spectral resolution for hyperspectral applications. Wang, Zhang, Du, Xia and Tao (2023) provide a systematic review of self-supervised EO pretraining, highlighting that no single recipe dominates across all downstream tasks.

A key limitation for urban socioeconomic regression is that these models expect multispectral input. The *CityLens* benchmark provides only RGB imagery—a common constraint in urban sensing where high-resolution commercial satellite imagery is available in natural colour but not in the six-band HLS configuration. Prior work has addressed this spectral mismatch through zero-padding, spectral interpolation, or feature-space projection, each imposing implicit assumptions about the relationship between visible and non-visible bands. We take a different approach: a learnable 1×1 convolutional channel adapter (Equation 5) that imposes no smoothness assumptions and learns an arbitrary linear mapping from RGB to the six-band input space during fine-tuning.

2.2. General Vision Models for Urban Analytics

Self-supervised Vision Transformers trained on natural photographs have emerged as surprisingly competitive with domain-specific EO models across remote sensing and urban analytics tasks. DINOv2 (Oquab et al., 2024) produces spatially rich patch-level features through self-distillation that generalise well across domains without domain-specific inductive biases. CLIP (Radford et al., 2021) aligns visual and textual representations for zero-shot transfer, though its image–text correspondence objective may limit utility for regression tasks depending on locally aggregated indicators. ResNet-50 (He et al., 2016) provides a strong convolutional baseline through local receptive fields and hierarchical feature aggregation.

Street-view imagery for socioeconomic inference has become a substantial subfield. Fan et al. (2023) show that street-view features encode "hidden city profiles"—latent representations correlated with economic and demographic indicators difficult to observe from overhead. Gebru et al. (2017) demonstrate that the visual appearance of streetscapes in Google Street View can predict neighbourhood demographics at census-block resolution, establishing complementarity between ground-level and satellite signals. In the satellite domain, Jean et al. (2016) combine satellite imagery with survey data for poverty prediction in sub-Saharan Africa via transfer learning from ImageNet-pretrained CNNs.

A common thread is that modalities are typically evaluated in isolation or combined without ensuring each modality's contribution is assessed on the same held-out locations. Our shared-split protocol addresses this gap, enabling the first controlled comparison of EO foundation model, general vision, and fusion branches on identical validation regions.

2.3. Parameter-Efficient Fine-Tuning

The small sample sizes in urban indicator datasets—*CityLens* provides hundreds, not millions, of training examples per task—severely constrain adaptation strategy. Full fine-tuning risks catastrophic forgetting: gradients required to fit a few hundred regression targets can overwrite spatial representations acquired from millions of satellite patches. LoRA (Hu et al., 2022) addresses this by injecting trainable low-rank matrices into frozen attention layers, reducing trainable parameters per layer from $O(d^2)$ to $O(2dr)$ where $r \ll d$ (Equation 7). The frozen weights preserve pretrained representations while low-rank perturbations steer attention toward task-relevant features. Hugging Face (2023) provide the PEFT toolkit used in this work.

The combination of LoRA with a channel adapter for adapting EO foundation models to RGB-only urban regression is, to our knowledge, novel. Prior LoRA applications in remote sensing focus on classification and segmentation with multispectral input; our setting tests whether parameter-efficient adaptation can bridge both a spectral mismatch and a domain shift simultaneously.

2.4. Urban Indicator Regression and the CityLens Benchmark

Liu et al. (2026) introduce *CityLens* as a benchmark for evaluating large vision-language models (LVLMs) on urban socioeconomic sensing across 17 cities spanning six continents (13 for globally available indicators). The original paper focuses on zero-shot LVLM evaluation using models such as GPT-4o and Gemini under three paradigms—direct metric prediction, normalised estimation, and feature-based regression—finding that even state-of-the-art LVLMs struggle with fine-grained, region-level indicator prediction. Their fine-tuned results (e.g., Qwen2.5-VL-7B achieving $R^2 = 0.872$ on building height and $R^2 = 0.628$ on GDP via direct metric prediction) establish upper-bound reference points for the benchmark but operate in a fundamentally different regime from the encoder-based regression we study here: fine-tuned LVLMs leverage billions of language-model parameters and are prompted with city identity and contextual text, whereas our approach isolates the contribution of visual representations alone.

This distinction is important for understanding modality contributions. The CityLens authors observe that street-view images alone often match or exceed satellite-plus-street configurations for LVLMs—a finding driven partly by the language models' ability to reason about ground-level scenes. Our encoder-based setting strips away this linguistic reasoning component, providing a cleaner test of what raw visual features from each modality contribute. Manvi, Khanna, Mai, Burke, Lobell and Ermon (2024) similarly operate in the language-model paradigm and do not address the paired satellite-and-street regression setting.

More broadly, prior work on urban indicator prediction evaluates modalities in isolation, on different data splits, or within modality-specific pipelines that preclude direct comparison. Evaluation protocols vary widely: some use random splits without controlling geographic proximity, others report single-seed results without variance estimates, and many compare modalities on non-overlapping subsets. To our knowledge, this is the first work to apply a shared-split paired evaluation protocol to *CityLens*, enabling unconfounded comparison with multi-seed uncertainty quantification.

3. Dataset: *CityLens*

3.1. Overview

CityLens (Liu et al., 2026) is a global, city-scale multi-task benchmark pairing satellite and street-view imagery for the same geographic locations across 13 cities on six continents. Distributed via Hugging Face (`Tianhui-Liu/CityLens-Data`), it supports both zero-shot LVLM evaluation—the original paper's focus—and learned regression, the setting we adopt here. Its geographic breadth is rare among urban prediction datasets, providing the distributional diversity needed to assess whether models generalise beyond the handful of well-studied cities in the United States and Western Europe that dominate existing benchmarks.

The dataset comprises four directories. **Benchmark/** contains one JSON task file per indicator, with each record providing the satellite image path, a list of street-view image paths (1–10 per location), the regression target, a normalised value, and geographic coordinates. **satellite_image/** stores one RGB PNG patch per location (256×256 px, ≈4.7 m/px, from Esri World Imagery), resized to 224×224 px for all experiments. **street_view_image/** contains ground-level RGB photographs sourced from Google Street View, Baidu Maps, or Mapillary depending on the city. **Dataset/** holds auxiliary files including the task registry and URL manifests.

3.2. Tasks

We evaluate on four globally available indicator tasks (Table 1), selected because ground-truth labels span all 13 cities, enabling genuine cross-regional evaluation. The `carbon` task is excluded pending a locked Prithvi reference run.

The tasks span a range of difficulty and signal modality. Building height has the strongest visual signal: vertical structure is directly observable from shadow geometry overhead and façade extent at ground level. GDP proxy—average GDP per grid cell in PPP 2005 USD—has diffuse visual correlates and a heavily right-skewed distribution. Healthcare accessibility (estimated walking time to the nearest facility) depends on infrastructure cues potentially more legible from the street than from above. Population density is a count proxy aggregated across heterogeneous footprints; as we show empirically, no branch achieves positive R^2 at this spatial scale.

3.3. Target Transform

All tasks apply a $\log(1+y)$ transform during training, with validation metrics computed in the raw target space after inversion. This is critical for GDP, whose heavy-tailed distribution would otherwise cause MSE loss to be dominated by extreme-value cities; it also stabilises gradient magnitudes for tasks spanning several orders of magnitude.

Table 1
CityLens tasks evaluated in this study. N denotes the street-available paired subset after coordinate-based path reconciliation (Section 3.4).

Task key	Target	Description	Scale	N (paired)
`gdp`	GDP proxy	Average GDP per grid cell (PPP 2005 USD). Heavy-tailed; $\log(1+y)$ transform applied during training.	Sat. patch	≈429
`acc2health`	Healthcare access	Estimated walking time (minutes) to the nearest healthcare facility.	Sat. patch	≈440
`build_height`	Building height	Mean net building height (metres) across the patch.	Sat. patch	≈398
`pop`	Population density	Total population count proxy per patch. Hardest task; all branches fail.	Sat. patch	≈402

3.4. Street-View Path Reconciliation

A critical preprocessing step was required before street-view experiments could proceed. The original benchmark JSON files referenced street-view images using legacy absolute paths from a different compute environment, while the distributed directory used a different filename namespace. The practical consequence was that zero valid street records were loaded despite images existing on disk—a silent failure that, without careful log inspection, would produce results computed on empty batches.

We resolved this via coordinate-based remapping: for each benchmark record, we located the nearest on-disk street-view image using geographic coordinates and rewrote the JSON with corrected relative paths. Post-reconciliation coverage reaches 86–87% of the satellite pool for all four tasks. The remaining locations lack matched street-view imagery, typically in cities with sparse ground-level coverage. This reconciliation step is itself a contribution to the practical usability of *CityLens* for learned regression.

3.5. Cross-Branch Comparability

The street-available subset (398–440 records) is a strict subset of the full satellite pool, so satellite-only results on the full pool and street/fusion results on the subset are not directly comparable. We address this in two ways. First, the shared-split protocol (Section 4.6) restricts all branches to the street-available subset using the same partition. Second, we clearly label any result using the full satellite pool, so readers can distinguish paired comparisons from indicative single-modality baselines.

4. Methods

4.1. Problem Formulation and Metrics

Notation. For task t, *CityLens* provides $\mathcal{D}_t = \left\{(\mathbf{x}_i^{\text{sat}},\ \{\mathbf{x}_{i,v}^{\text{str}}\}_{v=1}^{N_i},\ y_i)\right\}_{i=1}^{N}$, where $\mathbf{x}_i^{\text{sat}} \in \mathbb{R}^{3\times H\times W}$ is the satellite RGB patch, $\mathbf{x}_{i,v}^{\text{str}} \in \mathbb{R}^{3\times H\times W}$ is the v-th of up to $N_i \leq 10$ street-view images at location i, and $y_i \in \mathbb{R}^+$ is the regression target. For street-view experiments, we restrict to $\mathcal{D}_t^{\text{str}} \subseteq \mathcal{D}_t$ (Section 3.4). Each branch $b \in \{\text{sat, str, fus}\}$ produces $\hat{f}^{(b)}(\mathbf{x}_i) \in \mathbb{R}$ via a shared MLP regression head. This formulation deliberately isolates encoder and fusion choices from head capacity.

Target transform. We apply a $\log(1+y)$ transform to stabilise training on skewed targets:

$$\tilde{y}_i = \log(1 + y_i), \qquad \hat{y}_i = \exp\!\left(\hat{f}^{(b)}(\mathbf{x}_i)\right) - 1 \tag{1}$$

Training minimises MSE on transformed targets over the training split $\mathcal{T}_{t,s}$:

$$\mathcal{L}^{(b)} = \frac{1}{|\mathcal{T}_{t,s}|} \sum_{i\in\mathcal{T}_{t,s}} \left(\hat{f}^{(b)}(\mathbf{x}_i) - \tilde{y}_i\right)^2 \tag{2}$$

All evaluation metrics use raw-space predictions $\hat{y}_i$, ensuring errors are interpretable in original units (USD, minutes, metres).

Metrics. The coefficient of determination serves as the primary metric:

$$R^2(\hat{\mathbf{y}}, \mathbf{y}) = 1 - \frac{\sum_{i \in \mathcal{V}_{t,s}} (y_i - \hat{y}_i)^2}{\sum_{i \in \mathcal{V}_{t,s}} (y_i - \bar{y})^2}, \qquad \bar{y} = \frac{1}{|\mathcal{V}_{t,s}|} \sum_{j \in \mathcal{V}_{t,s}} y_j \tag{3}$$

We prefer R^2 over correlation because it penalises both bias and variance: 0 equals the constant-mean baseline, and negative values—frequent in our experiments—indicate predictions actively worse than that baseline. RMSE in raw units serves as the secondary metric:

$$\mathrm{RMSE}(\hat{\mathbf{y}}, \mathbf{y}) = \sqrt{\frac{1}{|\mathcal{V}_{t,s}|} \sum_{i \in \mathcal{V}_{t,s}} (y_i - \hat{y}_i)^2} \tag{4}$$

4.2. PRITHVI-EO-2.0+LORA Satellite Encoder

The adapted PRITHVI-EO-2.0 encoder (Figure 2) resolves a central tension: the backbone expects six spectral channels that do not exist in the input, yet its spatial representations are precisely what we wish to preserve.

Backbone. PRITHVI-EO-2.0-EO-2.0 is a ViT pretrained on HLS data via masked autoencoding, expecting 5-D tensors $(B, C{=}6, T, H, W)$ with channels aligned to Blue, Green, Red, NIR-narrow, SWIR-1, SWIR-2. The 3D patch embedding jointly tokenises spatial and temporal dimensions; we use $T{=}1$.

Channel adapter. We bridge the spectral gap with a learnable 1×1 convolution:

$$\mathbf{x}^{(6\mathrm{ch})} = \mathbf{W}_{1\times1} * \mathbf{x}^{\mathrm{sat}} + \mathbf{b}, \qquad \mathbf{W}_{1\times1} \in \mathbb{R}^{6\times3\times1\times1},\ \mathbf{b} \in \mathbb{R}^6 \tag{5}$$

Weights are initialised by uniformly broadcasting each RGB channel across two Prithvi bands, providing a spectrally neutral starting point. The adapter is intentionally minimal—24 trainable parameters—so its contribution is cleanly attributable. The output is reshaped to $(B, 6, T{=}1, H, W)$.

Forward pass. Combining adapter, backbone, and global average pooling (GAP) over patch tokens:

$$\mathbf{f}_{\mathrm{sat}} = \mathrm{GAP}\Big(\mathrm{Prithvi}_{\mathrm{LoRA}}\big(\mathrm{reshape}\big(\mathbf{W}_{1\times1} * \mathbf{x}^{\mathrm{sat}}\big)\big)\Big) \in \mathbb{R}^d \tag{6}$$

Only the adapter, LORA matrices, and MLP head are trainable; backbone parameters remain frozen.

LoRA fine-tuning. LORA (Hu et al., 2022) is applied to fused QKV and output projections in every transformer block. Each frozen $\mathbf{W}_0 \in \mathbb{R}^{d\times d}$ becomes:

$$\mathbf{W}' = \mathbf{W}_0 + \mathbf{B}\mathbf{A}, \qquad \mathbf{A} \in \mathbb{R}^{r\times d},\ \mathbf{B} \in \mathbb{R}^{d\times r},\ r \ll d \tag{7}$$

Only $\mathbf{A}$ and $\mathbf{B}$ receive gradients, yielding $2rd$ trainable parameters per layer versus d^2 for full fine-tuning—a compression that also acts as implicit regularisation, critical when each task provides only ≈350–400 training examples.

4.3. Satellite Baselines

ResNet-50 (He et al., 2016). An ImageNet-pretrained CNN whose convolutional inductive bias provides a baseline for patch-level encoding. It natively accepts 3-channel input. As reported in Section 5, it catastrophically fails on GDP ($R^2 \approx -4.0$).

DINOv2 ViT-B/14 (Oquab et al., 2024). A self-supervised ViT trained on curated web-scale data, serving as the primary non-EO transformer baseline. Performance gaps between DINOv2 and PRITHVI-EO-2.0 under our paired protocol can be attributed to pretraining domain differences, since both share the ViT architecture.

4.4. Street Encoder and Multi-View Pooling

Each location i is represented by up to $N_i \leq 10$ street-view photographs (Figure 3). A shared backbone encoder produces per-view embeddings:

$$\mathbf{e}_{i,v} = \mathrm{Enc}\Big(\mathbf{x}^{\mathrm{str}}_{i,v}\Big) \in \mathbb{R}^d, \qquad v = 1, \ldots, N_i \tag{8}$$

Mean pooling.

$$\mathbf{f}_{\text{str}}^{\text{mean}} = \frac{1}{N_i} \sum_{v=1}^{N_i} \mathbf{e}_{i,v} \tag{9}$$

Robust to variable view counts and parameter-free, but unable to distinguish informative from uninformative photographs.

Attention pooling. A learned relevance score produces softmax-normalised view weights:

$$\alpha_{i,v} = \frac{\exp\left(\mathbf{w}^\top \mathbf{e}_{i,v}\right)}{\sum_{v'=1}^{N_i} \exp\left(\mathbf{w}^\top \mathbf{e}_{i,v'}\right)}, \qquad \mathbf{f}_{\text{str}}^{\text{attn}} = \sum_{v=1}^{N_i} \alpha_{i,v}\, \mathbf{e}_{i,v} \tag{10}$$

Mean pooling is used in all main results as it proved equally effective and more stable. A two-layer MLP maps $\mathbf{f}_{\text{str}}$ to the scalar prediction. Available encoders are ResNet-50, CLIP ViT-B/16 (Radford et al., 2021), and DINOv2 ViT-B/14.

4.5. Fusion Architectures

All fusion experiments fix the satellite tower to PRITHVI-EO-2.0+LORA and vary the street encoder (Figure 1).

Late fusion. Feature concatenation followed by an MLP head:

$$\mathbf{z}^{\text{late}} = \left[\mathbf{f}_{\text{sat}};\, \mathbf{f}_{\text{str}}\right] \in \mathbb{R}^{2d}, \qquad \hat{y}_i = \text{MLP}\left(\mathbf{z}^{\text{late}}\right) \tag{11}$$

Gated fusion. A linear gate produces softmax-normalised modality weights:

$$d' = \min(d_{\text{sat}},\, d_{\text{str}}), \qquad (\alpha,\, \beta) = \text{softmax}\left(\mathbf{W}_g \left[\mathbf{f}_{\text{sat}};\, \mathbf{f}_{\text{str}}\right]\right), \quad \mathbf{W}_g \in \mathbb{R}^{2\times 2d} \tag{12}$$

$$\mathbf{z}^{\text{gate}} = \alpha\, \mathbf{f}_{\text{sat}}^{(d')} + \beta\, \mathbf{f}_{\text{str}}^{(d')}, \qquad \hat{y}_i = \text{MLP}\left(\mathbf{z}^{\text{gate}}\right) \tag{13}$$

where $\alpha + \beta = 1$. In principle, per-sample modality selection should suppress noisy inputs; in practice, as Section 5 shows, gated fusion does not consistently outperform late fusion.

4.6. Shared Validation Splits and Multi-Seed Protocol

Motivation. Without shared validation identifiers, R^2 gaps between modalities can be dominated by which cities appear in each branch's held-out set. This modality-split confound is not a hypothetical risk: we demonstrate in Section 5 that satellite PRITHVI-EO-2.0 drops from $R^2 = 0.390$ on the full pool to 0.042 on the paired split for healthcare access—a difference driven entirely by cohort composition.

Split generation. For task t, seed s, and validation fraction $v = 0.1$:

$$\left(\mathcal{T}_{t,s},\, \mathcal{V}_{t,s}\right) = \text{Split}\left(\mathcal{D}_t^{\text{str}},\ v{=}0.1,\ s\right), \qquad \frac{|\mathcal{V}_{t,s}|}{|\mathcal{D}_t^{\text{str}}|} = v \tag{14}$$

The split is persisted as a JSON file and loaded by all branches via a common flag, guaranteeing identical $\mathcal{V}_{t,s}$ across satellite, street, and fusion runs.

Multi-seed reporting. We report results for seeds $S = \{42, 43\}$ (14 paired training runs) with:

$$\mu_{t,b} = \frac{1}{|S|} \sum_{s\in S} R^{2\,(b)}_{t,s}, \qquad \sigma_{t,b} = \sqrt{\frac{1}{|S|-1} \sum_{s\in S} \left(R^{2\,(b)}_{t,s} - \mu_{t,b}\right)^2} \tag{15}$$

Two seeds suffice to detect sign flips and large instabilities, though we acknowledge this does not constitute a converged Monte Carlo study. Population density results are reported at seed 42 only, as all branches produce consistently negative R^2.

Table 2
Satellite-only results at seed 42. † = catastrophic failure; excluded from ranking. RMSE in raw target units.

Task	Model	Best ep.	R^2	RMSE	MAE
acc2health	DINOv2 ViT-B/14	20	0.099	11.611	8.775
	ResNet-50	22	0.212	10.853	7.204
	Prithvi+LoRA	9	**0.390**	**9.550**	**7.191**
build_height	DINOv2 ViT-B/14	18	0.679	3.954	2.812
	ResNet-50	26	0.800	3.118	2.281
	Prithvi+LoRA	11	**0.860**	**2.613**	**1.910**
gdp	DINOv2 ViT-B/14	19	0.454	3.78×10^8	2.33×10^8
	ResNet-50†	5	*−4.005*	1.15×10^9	3.93×10^8
	Prithvi+LoRA	14	**0.581**	3.31×10^8	1.98×10^8
pop	DINOv2 ViT-B/14	5	−0.184	23,176	11,363
	ResNet-50	2	−0.266	23,966	11,872
	Prithvi+LoRA	2	−0.032	21,641	10,020

4.7. Implementation Details

All runs use 224×224 px images and batch size 8, optimised with AdamW ($\lambda = 10^{-2}$). A dual learning-rate schedule assigns $\eta_{\text{adapt}} = 2\times10^{-4}$ to LoRA matrices and the channel adapter, and $\eta_{\text{head}} = 1\times10^{-3}$ to the MLP head—reflecting that the adapter operates near pretrained representations where large steps risk disruption, while the randomly initialised head benefits from faster convergence. The Prithvi-EO-2.0 backbone is frozen. Early stopping monitors validation R^2 with per-task epoch budgets: 20 for GDP, 30 for healthcare and building height, 5 for population density. Frameworks: PyTorch, `timm` (Wightman, 2019), TerraTorch, and HuggingFace PEFT (Hugging Face, 2023). Memory requirements: ≈3–5 GB (satellite), ≈11.5 GB (street), ≈24 GB (fusion).

5. Results

This section presents the full quantitative evaluation, organised by modality branch. Bold denotes the highest R^2 per task among valid baselines; † marks catastrophic failures excluded from rankings. Cross-branch ordering claims are based exclusively on the paired results in Table 5; single-seed results are descriptive.

5.1. Satellite-Only Results (Seed 42)

Prithvi-EO-2.0+LoRA leads on every task, with substantial margins. On building height ($R^2 = 0.860$), it outperforms ResNet-50 by +0.060 and DINOv2 by +0.181. This gap is notable given that DINOv2 is also a ViT pretrained at scale; the difference suggests EO-specific spatial priors encode urban morphological features—shadow geometry, roofline patterns, street-grid regularity—that natural-image pretraining does not capture. On GDP ($R^2 = 0.581$ vs. DINOv2's 0.454), the advantage persists while ResNet-50 catastrophically fails ($R^2 \approx -4.0$; analysed as F1 in Section 6.5). Healthcare access shows the widest margin: Prithvi-EO-2.0 at 0.390 exceeds DINOv2 by +0.291. Population density is the outlier: all encoders produce negative R^2, indicating the problem lies in signal content rather than model capacity.

5.2. Street-Only Results (Seed 42)

The street results reveal a strikingly different encoder hierarchy. ResNet-50 dominates GDP, healthcare, and building height, substantially outperforming both ViT-based alternatives. This reversal—ResNet-50 is the weakest non-catastrophic satellite encoder but the strongest street encoder—suggests that convolutional architectures' local receptive fields and hierarchical aggregation may be better suited to fine-grained street-level cues (façade materials, signage, road surface quality) than the global attention patterns of ViTs pretrained on single-image classification.

The satellite–street gap varies meaningfully across tasks. For healthcare (satellite 0.390 vs. street 0.330), the gap is narrow, consistent with ground-level infrastructure cues providing complementary signal. For building height (0.860 vs. 0.345), overhead geometric signal dominates. These task-specific gaps foreshadow the fusion results. Population density remains intractable: DINOv2's +0.006 is within the noise floor.

Table 3
Street-only results at seed 42 (street-available subset). Best encoder per task in bold.

Task	Encoder	Best ep.	R^2	RMSE	MAE
gdp	CLIP ViT-B/16	19	−0.029	5.75×10^8	3.39×10^8
	DINOv2 ViT-B/14	13	0.034	5.57×10^8	3.44×10^8
	ResNet-50	7	**0.363**	4.53×10^8	3.09×10^8
acc2health	CLIP ViT-B/16	3	−0.003	10.962	7.365
	DINOv2 ViT-B/14	11	0.010	10.890	7.447
	ResNet-50	21	**0.330**	**8.960**	**6.308**
build_height	CLIP ViT-B/16	17	0.234	6.192	4.916
	DINOv2 ViT-B/14	22	0.337	5.760	4.636
	ResNet-50	10	**0.345**	**5.726**	**4.524**
pop	ResNet-50	2	−0.435	14,828	9,242
	CLIP ViT-B/16	5	−0.306	14,143	9,082
	DINOv2	2	**0.006**	**12,341**	**9,162**

Table 4
Fusion results at seed 42 (PRITHVI-EO-2.0+LORA satellite tower + street encoder).

Task	Street encoder	Type	Best ep.	R^2	RMSE
acc2health	ResNet-50	Gated	3	0.161	10.024
	ResNet-50	Late	12	**0.191**	**9.845**
build_height	DINOv2 ViT-B/14	Gated	21	0.865	2.596
	DINOv2 ViT-B/14	Late	23	**0.872**	**2.527**
gdp	ResNet-50	Late	6	0.371	4.50×10^8
	ResNet-50	Gated	20	**0.444**	4.23×10^8
pop	ResNet-50	Gated	4	−0.294	14,080
	ResNet-50	Late	4	−0.349	14,374

5.3. Fusion Results (Seed 42)

The fusion results challenge the common assumption that combining modalities uniformly improves performance. Building height is the sole task where fusion exceeds the best unimodal branch: late fusion achieves $R^2 = 0.872$ versus satellite-only 0.860. On GDP and healthcare, fusion substantially underperforms satellite-only PRITHVI-EO-2.0 (0.444 vs. 0.581; 0.191 vs. 0.390), indicating the street stream introduces noise that the fusion head cannot suppress. Notably, the gated fusion mechanism designed for modality selection fails to rescue these tasks—on healthcare, gated fusion converges at epoch 3, suggesting gate collapse to a fixed ratio.

Late fusion consistently matches or exceeds gated fusion, suggesting that the gate's additional parameterisation requires more training signal than these small datasets provide.

5.4. Paired Multi-Seed Results

Table 5 presents the central results: paired, shared-split analysis across two seeds. These are the only results supporting cross-branch ordering claims.

Building height: the cleanest result. Fusion leads with $\mu = 0.850$ ($\sigma = 0.032$), satellite follows at $\mu = 0.811$ ($\sigma = 0.083$), street at $\mu = 0.342$ ($\sigma = 0.171$). The ordering is preserved at both seeds, and fusion's lower variance than satellite alone (0.032 vs. 0.083) suggests that combining modalities not only improves the point estimate but also stabilises predictions.

Healthcare access: directional but unreliable. The mean ordering favours street over satellite (0.340 vs. 0.042), *reversing* the seed-42 single-seed ranking where satellite appeared to lead. This reversal concretely demonstrates the

Table 5
Paired shared-split two-seed results ($s \in \{42, 43\}$). ✓ = stable across seeds; ≈ = directional, high variance; × = not supported; † = sign flip.

Task	Branch	R^2 (mean ± std)	RMSE (mean ± std)	Status
`build_height`	**Fusion (late, DINOv2)**	$\mathbf{0.850 \pm 0.032}$	2.612 ± 0.119	✓
	Satellite (PRITHVI-EO-2.0)	0.811 ± 0.083	2.926 ± 0.836	✓
	Street (ResNet-50)	0.342 ± 0.171	5.492 ± 1.061	✓
`acc2health`	**Street (ResNet-50)**	0.340 ± 0.400	9.755 ± 4.290	≈
	Satellite (PRITHVI-EO-2.0)	0.042 ± 0.018	11.809 ± 1.654	✓
`gdp`	Satellite (PRITHVI-EO-2.0)	0.192 ± 0.195	4.96×10^8	†
	Street (ResNet-50)	0.119 ± 0.497	5.05×10^8	†
`pop`	All branches	seed-42 only	—	×

Table 6
Cross-branch summary. Seed-42 best R^2 per modality; "Evidence" references the paired analysis.

Task	Satellite	Street	Fusion	Leader (evidence)
GDP	0.581	0.363	0.444	Satellite (seed-42; sign-flip at s=43)
Healthcare	0.390	0.330	0.191	Satellite (marginal; paired uncertain)
Build. height	0.860	0.345	**0.872**	Fusion ✓ (μ=0.850, σ=0.032)
Population	−0.032	0.006	−0.294	None meaningful

modality-split confound: satellite PRITHVI-EO-2.0's unpaired R^2 of 0.390 drops to 0.042 on the paired split, a difference driven by cohort composition rather than model quality. However, the street branch's $\sigma = 0.400$ exceeds its mean ($\sigma/\mu = 1.18$), so the advantage cannot be claimed as robust.

GDP: sign flip. The street branch produces $R^2 = +0.47$ at seed 42 but -0.23 at seed 43—a sign flip turning a seemingly viable modality into one worse than the constant-mean predictor. Had we reported only seed 42, the street branch would have appeared promising; the second seed reveals this as a statistical artefact. We do not claim a robust modality winner for GDP.

5.5. Cross-Branch Summary

The pattern is one of task-dependent modality value: no single branch dominates, fusion helps only where modalities provide genuinely complementary signals, and seed-42 rankings are not always stable under multi-seed evaluation.

6. Discussion

6.1. Why PRITHVI-EO-2.0 Leads Despite RGB Input

The consistent superiority of PRITHVI-EO-2.0+LORA over all satellite baselines despite receiving only three of six expected spectral channels is the most striking encoder-level finding. Two complementary mechanisms explain this result.

First, large-scale EO pretraining encodes spatially meaningful features that transfer to urban regression even under spectral mismatch. The masked autoencoder objective rewards learning of geometric regularities—street-grid periodicity, roofline patterns, shadow angles, land-cover textures—that are encoded in the frozen weights and available downstream regardless of input spectral channels. The channel adapter need only project RGB into a six-dimensional space that activates these features; it does not need to hallucinate genuine NIR or SWIR information.

Second, LORA fine-tuning adapts transformer layers without overwriting EO priors. Full fine-tuning on ≈400 examples per task would risk catastrophic forgetting, reducing PRITHVI-EO-2.0 to a randomly initialised ViT on few hundred samples. The low-rank constraint acts as implicit regularisation, preserving pretrained attention while allowing targeted adjustments.

The consistent margin over DINOv2—which also uses a ViT architecture and large-scale pretraining—ranges from +0.127 on GDP to +0.291 on healthcare at seed 42. Since both models share architecture and are adapted with similar regimes, this gap can be attributed to pretraining domain: EO pretraining provides spatial features that natural-image pretraining does not, even when downstream input is limited to overlapping RGB bands. This finding carries practical implications: investing in large-scale EO pretraining pays dividends even when full spectral data is unavailable at deployment.

6.2. Relationship to CityLens LVLM Results

Our encoder-based results complement the LVLM evaluation in the original *CityLens* paper (Liu et al., 2026) and provide several cross-paradigm insights. On building height, our best fusion result ($R^2 = 0.872$ at seed 42) closely matches the fine-tuned Qwen2.5-VL-7B ($R^2 = 0.872$) and substantially outperforms all zero-shot LVLMs evaluated in the feature-based regression paradigm, where the best model achieves $R^2 = 0.807$ (UrbanVLP). This convergence across fundamentally different paradigms—learned regression with frozen encoders versus fine-tuned billion-parameter language models—suggests that building height prediction may be approaching a performance ceiling for this dataset at the available spatial resolution.

For GDP, our satellite-only PRITHVI-EO-2.0 ($R^2 = 0.581$) is competitive with the fine-tuned Qwen2.5-VL-7B ($R^2 = 0.628$), despite having orders of magnitude fewer trainable parameters and no access to textual city-identity prompts. This narrows the gap between encoder-based and LVLM-based approaches and suggests that for macro-scale spatial patterns, visual representations alone carry most of the predictive signal.

A notable divergence concerns the relative value of street-view imagery. Liu et al. (2026) find that street-view images alone often match combined satellite-plus-street configurations for LVLMs, attributed partly to language models' ability to reason about ground-level scenes. Our encoder-based setting—which strips away linguistic reasoning—reverses this pattern: satellite imagery dominates most tasks. This suggests that the street-view advantage observed in LVLM evaluations may reflect language-model priors rather than the raw visual information content of ground-level photographs.

6.3. When Fusion Helps and When It Does Not

The task-dependent value of fusion is the central empirical theme. Building height is the one task where fusion demonstrably helps, and the mechanism is interpretable: street-view photographs encode vertical structure—storey counts, fenestration patterns, architectural style—that satellite imagery captures only indirectly through shadow length and footprint area. The paired evidence confirms stability ($\mu = 0.850$, $\sigma = 0.032$) and shows that fusion variance is actually lower than satellite-only variance (0.083), suggesting partially orthogonal error sources whose combination stabilises predictions.

For GDP and healthcare, fusion actively degrades performance (GDP: 0.444 vs. satellite-only 0.581; healthcare: 0.191 vs. 0.390). Three factors likely contribute. First, these tasks depend on macro-scale patterns—land-use composition, infrastructure density, neighbourhood morphology—better captured from a single overhead vantage than from street photographs sampling a narrow spatial slice. Second, street-view features may encode location-specific noise (parked cars, pedestrians, transient signage) uncorrelated with the target, diluting the satellite signal upon concatenation. Third, the doubled input dimensionality of fusion heads demands more training signal for effective feature selection—particularly acute on datasets of a few hundred examples.

The underperformance of gated fusion relative to late fusion across all tasks challenges the intuition that explicit modality selection helps when one modality is noisy. The gate's early convergence on healthcare (epoch 3) is diagnostic of gate collapse: with insufficient training signal, the gate degenerates to a fixed blending coefficient, losing the sample-dependent modulation it was designed to provide.

6.4. The Population Density Problem

Population density's consistent failure across all branches—no configuration achieves $R^2 > 0$ except DINOv2 street's negligible +0.006—indicates a fundamental observability limit rather than a modelling deficiency. The core difficulty is the many-to-one mapping between built form and occupancy: a dense high-rise district and a dense informal settlement may appear visually similar overhead yet differ by an order of magnitude in population due to differences in floor-area ratio and occupancy patterns invisible at this spatial resolution and spectral range. The *CityLens* population target itself is a count proxy aggregated across heterogeneous footprints, potentially introducing patch-level label noise. Auxiliary data—nighttime radiance, mobile phone traces, or building footprint databases with storey counts—may be required for positive R^2.

6.5. Structured Failure Documentation

We document five failure modes, reported to make evaluation more transparent.

F1 — Catastrophic satellite baseline (GDP + ResNet-50). $R^2 \approx -4.0$, with RMSE three times PRITHVI-EO-2.0's. The heavy-tailed GDP distribution means MSE is dominated by extreme-value cities; ResNet-50's local receptive field may lack the global context needed to handle these outliers. The shared learning-rate schedule (designed for ViTs) may also be poorly suited to convolutional optimisation landscapes.

F2 — Population across all branches. A task-level failure rooted in the observability limit of RGB imagery for population estimation, not a modelling deficiency.

F3 — Fusion degrades GDP and healthcare. The best fusion R^2 falls below satellite-only PRITHVI-EO-2.0 for both tasks, directly contradicting the assumption that multimodal data always helps or at worst does not harm.

F4 — Gated fusion underperforms late fusion. The more complex mechanism fails to outperform simple concatenation, with gate collapse evident from early convergence on healthcare.

F5 — Cross-branch comparability on unpaired runs. Satellite PRITHVI-EO-2.0 drops from $R^2 = 0.390$ (full pool) to 0.042 (paired split) on healthcare—a difference attributable to cohort composition, illustrating the modality-split confound in action.

6.6. Implications for EO Foundation Model Benchmarking

The modality-split confound is not unique to *CityLens*: any multimodal benchmark allowing different branches to draw from different validation pools is susceptible. The GDP sign flip demonstrates that single-seed reporting can produce conclusions that are directionally wrong, not merely imprecise. These affect model selection in operational settings where the "best" published model is deployed without further validation.

We advocate three minimum standards for future EO evaluations: (i) shared validation identifiers across all branches, implemented via distributable split files; (ii) multi-seed reporting with at least two seeds, ideally three or more for high-variance tasks; (iii) explicit documentation of modality-available subsets used for paired comparisons, enabling assessment of effective sample size and geographic composition.

6.7. Limitations

Spatial autocorrelation. Random ID-based splits may allow spatially proximate training and validation records, inflating R^2 via label smoothness. Geographic k-fold or city-held-out evaluation (Jean et al., 2016) would be more conservative.

Adapter alternatives. The 1×1 adapter learns an unconstrained mapping; comparison against zero-padding, spectral interpolation, or identity-mapping baselines would isolate its contribution.

Two seeds. Sufficient to detect sign flips but not a full Monte Carlo study. Three or more seeds would tighten variance estimates, particularly for healthcare and GDP.

Fairness. Models may systematically underperform in cities historically under-represented in satellite products or with different target distributions. Per-city error analysis—examining whether prediction quality correlates with income level or imagery availability—is a prerequisite for deployment beyond benchmarking. The CityLens authors' own bias audit reveals substantial performance disparities between Global North and Global South cities (Liu et al., 2026), a concern our cross-regional results do not yet address at the per-city level.

7. Conclusions

This paper has presented a paired, modality-aligned evaluation of PRITHVI-EO-2.0-EO-2.0 and generic vision encoders on four *CityLens* urban indicator regression tasks. The central methodological contribution—a shared validation split protocol fixing held-out region identifiers across satellite, street-view, and fusion branches—eliminates the modality-split confound that undermines cross-modal comparison in prior work.

Three empirical findings emerge. First, PRITHVI-EO-2.0+LORA achieves the highest R^2 among satellite encoders on every task despite receiving only RGB input, with consistent margins over DINOv2 (+0.127 to +0.291) attributable to EO-specific spatial priors that transfer usefully even under spectral mismatch. Second, multimodal fusion benefits building height only ($\mu = 0.850$, $\sigma = 0.032$), where street-level façade geometry complements overhead shadow-and-footprint signal; for GDP and healthcare, fusion actively degrades performance. Third, the GDP sign flip—street-branch R^2 swinging from +0.47 to −0.23 across two seeds—demonstrates that single-seed evaluation can produce directionally wrong conclusions.

These findings carry two practical messages. For the growing community adapting EO foundation models to downstream tasks with limited spectral data: EO pretraining pays dividends even when only RGB is available, and lightweight adaptation (channel adapter + LoRA) suffices to unlock this value. For the evaluation community: shared splits, multi-seed reporting, and structured failure documentation should be minimum standards for any multimodal benchmark, not optional embellishments.

Several extensions would strengthen these findings. Geographic k-fold evaluation, holding out entire cities, would address spatial autocorrelation and test generalisation across urban typologies. Per-city fairness metrics would assess whether prediction quality correlates with city income or imagery availability. Temporal PRITHVI-EO-2.0 inputs would leverage the backbone's multi-date pretraining capabilities unused in our single-frame setup. Additional seeds would tighten variance bounds for high-variance tasks. Extension to remaining *CityLens* targets—including carbon footprint—would test whether the observed patterns generalise across a wider set of urban indicators.

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Data availability

The *CityLens* dataset is publicly available at https://huggingface.co/datasets/Tianhui-Liu/CityLens-Data (Liu et al., 2026). The shared split JSON files, training configuration files, experiment logs, and evaluation scripts supporting all results are deposited at [repository URL upon acceptance] under a CC BY 4.0 licence. All results are reproducible using the hyperparameters and seeds documented in Section 4.7 and the supplementary material.

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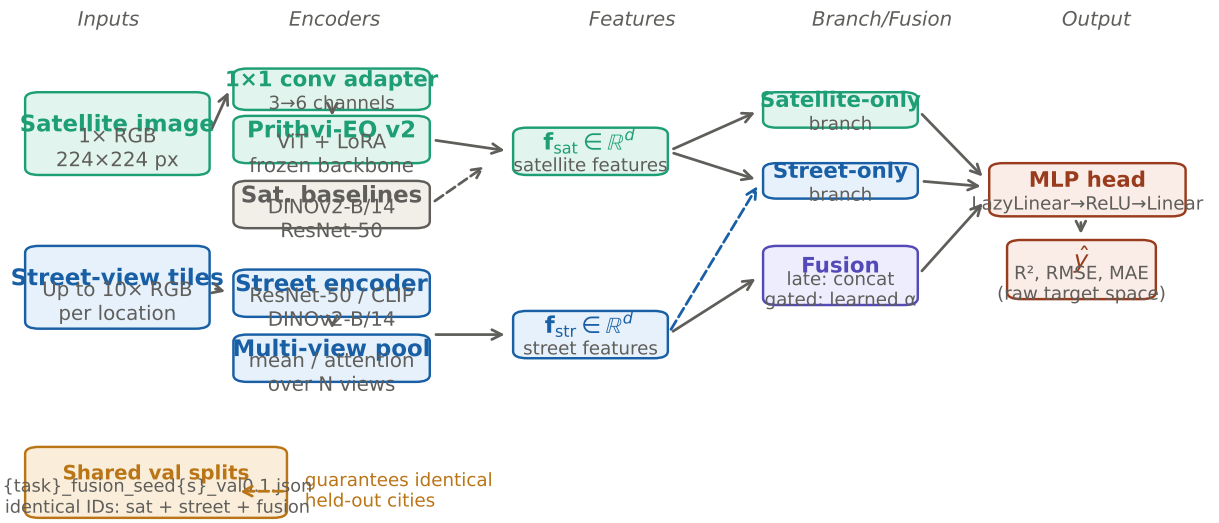

Figure 1: Overview of the *CityLens* multimodal pipeline. Satellite RGB patches and street-view tiles are processed by separate encoder towers. Features are routed to one of three branches: satellite-only, street-only, or fusion (late concatenation or gated weighting), each sharing a common MLP regression head. A single shared split JSON file fixes identical validation region identifiers across all three branches, eliminating the modality-split confound.

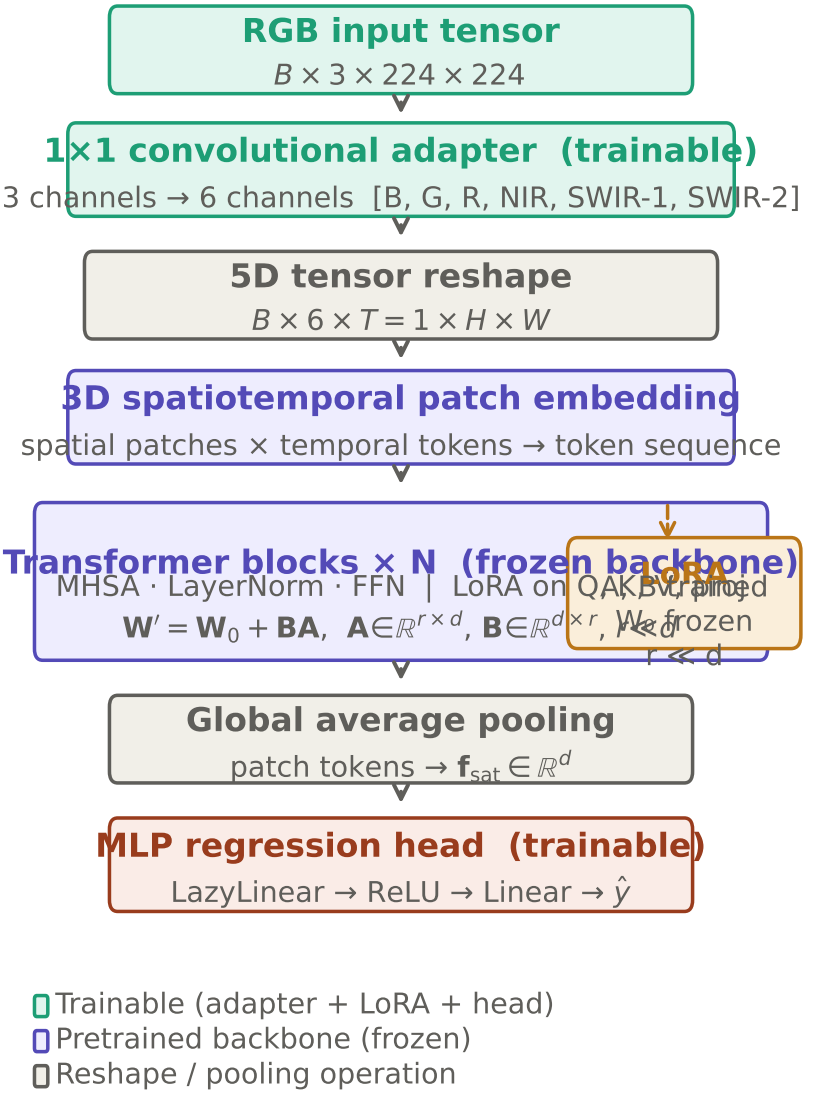

Figure 2: Architecture of the PRITHVI-EO-2.0-EO-2.0 satellite encoder with LORA adaptation. A learnable 1×1 convolutional adapter maps the 3-channel RGB input to 6 channels aligned with PRITHVI-EO-2.0's HLS band order. The backbone weights $\mathbf{W}_0$ are frozen throughout training; only the adapter, the LORA matrices $\mathbf{A}$ and $\mathbf{B}$ (Equation 7), and the MLP regression head are updated. Colour coding: teal = trainable components; purple = frozen pretrained backbone; grey = reshape and pooling operations.

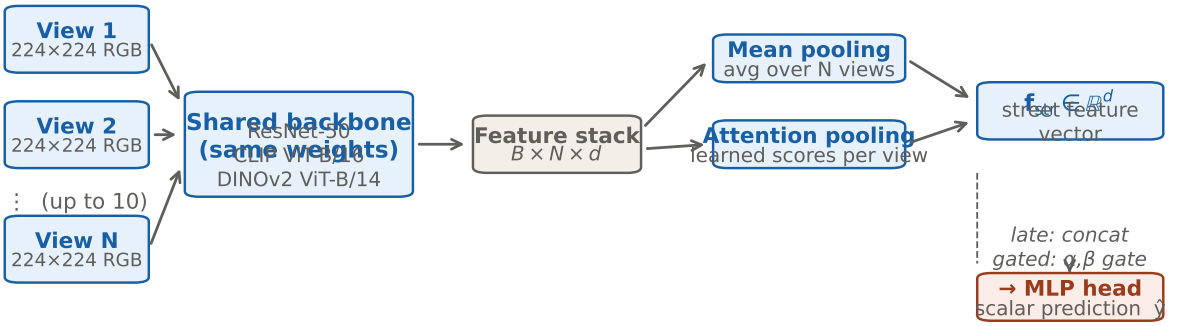

Figure 3: Street encoder and multi-view pooling. Up to 10 street-view images per location are processed by a shared backbone (ResNet-50, CLIP ViT-B/16, or DINOv2 ViT-B/14), producing a per-view feature stack of shape $B \times N \times d$. Mean pooling averages across views; attention pooling learns a scalar relevance score per view via a single linear projection. The pooled vector $\mathbf{f}_{\text{str}} \in \mathbb{R}^d$ is passed to the MLP head for street-only prediction, or concatenated with $\mathbf{f}_{\text{sat}}$ for fusion.

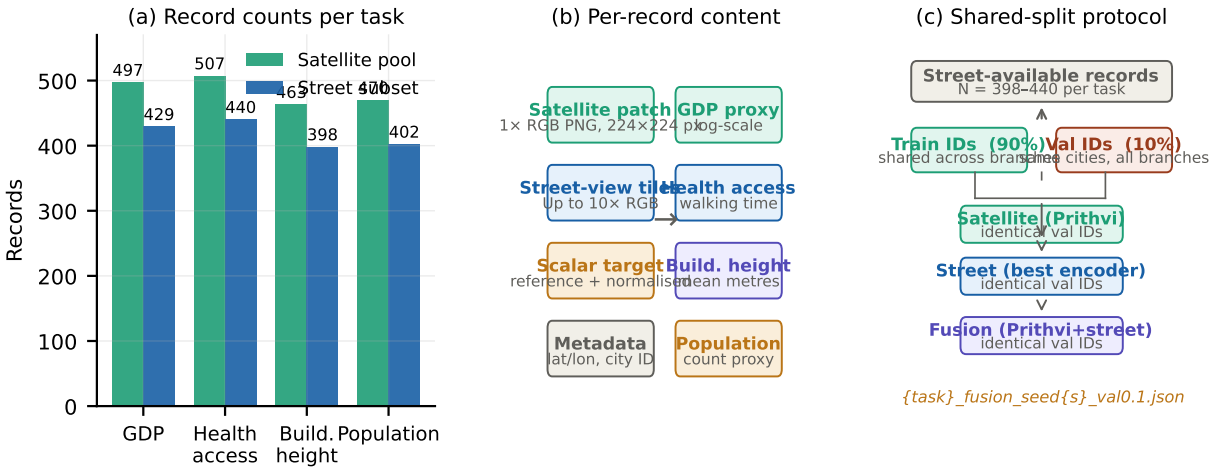

Figure 4: CityLens dataset overview. (a) Record counts per task, showing the full satellite pool and the street-available subset after coordinate-based path reconciliation (86–87% coverage). (b) Per-record content structure: each record contains one satellite patch, up to 10 street-view tiles, a scalar target, and latitude/longitude metadata. (c) Shared-split evaluation protocol: a single JSON split file fixes identical validation region identifiers (Val IDs, 10%) across all three branches (satellite, street, fusion), with the remaining 90% used for training.

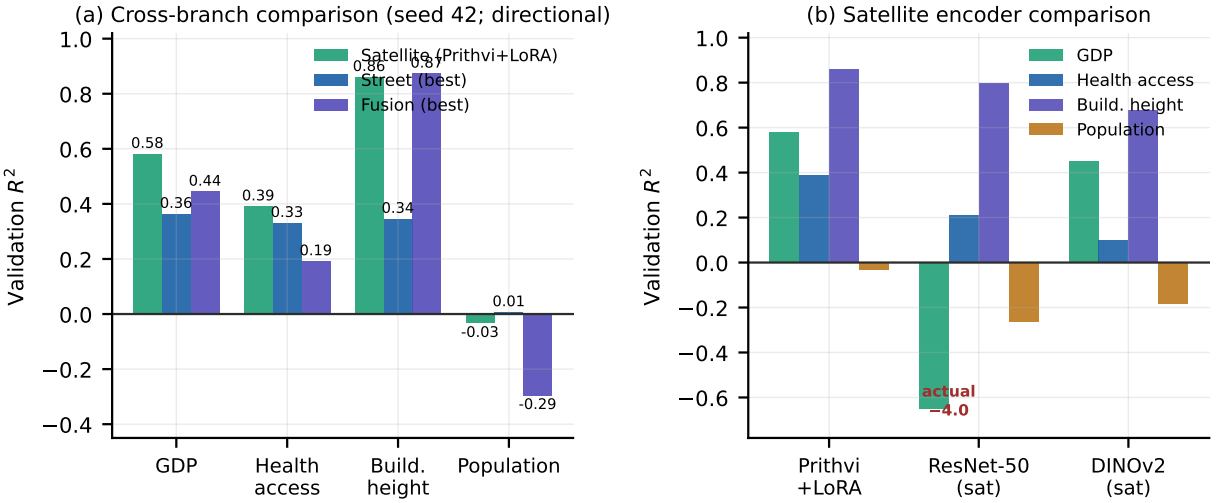

Figure 5: Seed-42 quantitative results. (a) Cross-branch comparison showing the best R^2 per task for satellite (Prithvi+LoRA), street (best encoder per task), and fusion (best configuration per task). Cross-branch ordering is indicative only at a single seed; see Table 5 for paired multi-seed evidence. (b) Satellite encoder comparison across all four tasks. The ResNet-50 bar for GDP is clipped at -0.65; the actual value is $R^2 \approx -4.0$ (catastrophic failure; Section 6.5). Error bars are omitted for seed-42-only results; mean $\pm$ std values are reported in Table 5.